



that time. It's all well and good to see proof of your love for your girlfriend carved on a tree years in the future, but it can get embarrassing with your wife standing next to you.

The biggest threat our online identities face is the damage that can be caused by unscrupulous and unauthorized intrusion. Many of us have been through the embarrassment of emails sent with our good name to unsuspecting habitants of our address books, and the sheer horror that follows. The messy explanations, "No, I'm NOT selling 'enhancement' products" OR "No, I don't have any relatives in Nigeria".

The worrisome fact is that this is so very easy in the digital world. Even though new forms of communication have become more popularised, and better secured, a large portion of our online persona lies in our email boxes.

Stories may be spun and rumours may fly, but when it comes right down to it, in the

authorship of documents with his signature.

A signature is something that anyone can read and possibly verify, and as such it gives no security to the contents of the documents. Like any form you fill (in quadruplicate) for a government organisation, your signature just verifies that you filled the form, the contents of it are still open for perusal by the dozens of people it bounces between before reaching its destination or the Recycle Bin.

Any signature scheme needs to enable anyone to 1) easily and effectively sign their own documents 2) easily verifiable and identifiable as yours (or anyone else's), and 3) be very difficult if not impossible to forge. We see this in our day-to-day signatures, since they are almost a part of our identity, they are easy to produce. They can be verified by matching with a copy, and it is difficult to produce someone else's signature. A digital signature by comparison provides much

"It's not who you are underneath, but what you do that defines you"

A Secure Digital Identity

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It's true – while your greatest accomplishments may barely see the light of day, it's your most embarrassing moments that will most often reach a wider audience. While we try our best to live our lives in frivolous anonymity, we are on some level aware, that we leave clues to our identity as we move through our lives. We may call up work and say we are sick with a fever while we watch a film, all the while knowing that all it takes is a simple call home to confirm our dishonesty.

A digital identity

As the internet draws inroads deeper into our personal lives, we find ourselves increasingly concerned about our online personality.

Today, a simple Google search can yield all your tweets on Twitter, all your blog posts, your forum interactions, your video posts, the list goes on. All the breadcrumbs Hansel and Gretel need to reach the deepest recesses of your identity.

One rarely worries about the implications of what we leave behind, and often forget who all can see it. The ease and seeming anonymity of the internet are just too lucrative. What many people do online is the equivalent of carving every bench, every tree, every wall we encounter in our lives with our thoughts at

world outside computers, it is your signature that defines the authenticity of the written word. Flawed as it might be it is a system that provides a measure of security to our transactions.

Digitised validity

How would you share a secret? Perhaps by whispering it to a friend, or discreetly slipping a note of the same. However, in most situations neither is an option. We may need to get secretive or sensitive information across, however the party delivering the message cannot be trusted with it. When we look at computers transmitting information over the internet, what we have is a communication being sent over a possibly unsecured channel that could be monitored or even tampered by adversaries. Depending on the content of the message, and its target audience, we relate to different security schemes. In some cases verifiability of a message's authenticity is all that is required, while in other cases the authenticity of the source is more important.

Here lies the difference between the concepts of digital signatures, message authenticity verification and encryption.

Digital signatures

A digital signature may be taken pretty much in the same terms as a physical signature, it identifies the authorship of a document, thus authenticating the source. It gives security to the author, as he is only liable to accept

more security than an ink signature as it is practically unforgeable.

Authenticity verification

Authenticity verification is more to do with ensuring that the contents of the document are safe from modification, and any modification made by unscrupulous entities can easily be detected. This is similar to laminating a paper document with a plastic sheet, this prevents modifications, and any forced modifications made are liable to be easily detected.

Here, the source of the document is not important, what is important is the validity of the document. Although such is the case with the example of lamination, it is not necessary everyone should be able to detect unauthorised modifications. We can take the example of a document written in special ink which cannot be differentiated from normal ink by most people. In such a case, only people with the ability to differentiate the special ink will be able to detect whether there has been any unauthorised modifications.

Again as in the case of digital signatures, authentication tags need to be easy to produce and verified for the parties involved, but impossible for an outsider to produce.

Encryption

Encryption is the most effective way to keep things away from prying eyes. Unlike authentication tags and digital signatures,